

CHANGE REQUEST						CR-Form-v12.2
23.228	CR	1597	rev	-	Current version:	19.1.0

For **HELP** on using this form: comprehensive instructions can be found at
<http://www.3gpp.org/Change-Requests>.

Proposed change affects: UICC apps ☐ ME ☒ Radio Access Network ☐ Core Network ☒

Title:	KI#7: Updates on determination and handling of multiplexing		
Source to WG:	Samsung		
Source to TSG:	SA2		
Work item code:	NG_RTC_Ph2	Date:	2025-02-10
Category:	C		
Use <u>one</u> of the following categories:			
F (correction)			
A (mirror corresponding to a change in an earlier release)			
B (addition of feature),			
C (functional modification of feature)			
D (editorial modification)			
Detailed explanations of the above categories can be found in 3GPP TR 21.900 .			
Release: Rel-19			
Use <u>one</u> of the following releases:			
Rel-8 (Release 8)			
Rel-9 (Release 9)			
Rel-10 (Release 10)			
Rel-11 (Release 11)			
...			
Rel-16 (Release 16)			
Rel-17 (Release 17)			
Rel-18 (Release 18)			
Rel-19 (Release 19)			

Reason for change:	How the IMS AS and the DCSF determine whether the data channel streams are multiplexed or de-multiplexed need to be described in accordance to the Nimsas_MediaControl_MediaInstruction service of IMS AS, Nmf_MRM services of MF, and SDP offer sent by the UE.		
Summary of change:	- How the UE requests the multiplexing of data channel streams in SDP offer is clarified. - The determination and handling of multiplexed or de-multiplexed data channel streams at DCSF, IMS AS are clarified. - The case of multiplexing the local bootstrap data channels and the remote bootstrap data channels is clarified as well.		
Consequences if not approved:	Incomplete specification.		

Clauses affected:	AC.7.10.1, AC.7.10.3		
Other specs affected: (show related CRs)	Y	N	
		X	Other core specifications
		X	Test specifications
		X	O&M Specifications
Other comments:	The details of IMS AS services and MF services are aligned with CR1586.		
This CR's revision history:			

AC.7.10.1 General

When multiple data channels are used in an IMS session, the UE and IMS network may support SDP negotiation to multiplex multiple data channel streams into a single data channel SDP media description m line and transport streams of multiple applications in the same SCTP connection, to save the number of m lines used by the UE and to improve the efficient usage of media resources.

The UE and IMS network also need to support data channel de-multiplexing when supporting data channel multiplexing. If the UE receives streams targeting to different applications over a single SCTP connection, it identifies the applications and delivers the data of the streams to corresponding applications. If the IMS network receives multiplexed streams which have different remote endpoints, e.g. a local bootstrap data channel and a remote bootstrap data channel of a UE, the MF terminates the multiplexing and sends different streams to corresponding endpoints based on instructions of DCSF.

The following multiplexing scenarios are supported in this Release:

- Multiplexing local bootstrap data channel and remote bootstrap data channel between the UE and its home IMS network using a single SDP m line media description if both support data channel multiplexing capability;
- Multiplexing different application data channels for applications with compatible QoS requirements between the UE and its IMS network using a single SDP m line media description if both support multiplexing capability;
- Multiplexing different application data channels for applications with compatible QoS requirements between the originating IMS network and terminating IMS network using a single SDP m line media description if both support DC multiplexing capability.

Multiplexing the data channels using the same stream ID are not supported.

The following de-multiplexing scenarios are supported in this Release:

- The supporting IMS network de-multiplexes local bootstrap data channel and remote bootstrap data channel from a SDP media description towards a UE that does not support multiplexing;
- The supporting IMS network de-multiplexes application data channels from a SDP media description, if the data channels have different endpoints e.g. a P2P application and a P2A application or towards a UE that does not support multiplexing;
- The supporting originating IMS network de-multiplex data channels towards remote network if the remote network or remote UE does not support DC multiplexing.

If the UE decides to use data channel multiplexing, when the UE generates SDP offer in INVITE or re-INVITE request, the DC stream IDs associated with the data channels to be multiplexed and the binding information combined with the data channels are included in the SDP offer for a single m line.

NOTE: How the application id and the DC stream IDs are included in the SDP is specified in SA WG4.

For multiplexing local bootstrap data channel and remote bootstrap data channel scenario, the UE includes multiple data channel application sources (e.g. a=dcmap:0, a=dcmap:10, a=dcmap:100, and/or a=dcmap:110) in a single m line of the SDP offer in (re-)INVITE request.

For multiplexing different application data channels scenario, the UE includes multiple application data channel streams (e.g. more than one a=dcmap lines with different a=3gpp-req-app lines) followed by a bootstrap data channel stream in a single m line in the SDP offer in (re-)INVITE request.

The IMS AS notifies the event of data channel multiplexing along with the data channel media information and binding information to the DCSF if data channel multiplexing is allowed for the subscriber.

The DCSF determines whether and how the data channels are multiplexed. When the DCSF determines that some data channels are allowed to be multiplexed, it instructs the IMS AS to multiplex the specific data channel streams to the same m line on the MF. The IMS AS further instructs MF to reserve a single media termination for the streams to be multiplexed.

If the DCSF determines that a data channel needs to be de-multiplexed from a SCTP connection, it instructs the IMS AS to de-multiplex the specific data channel stream on the MF. Based on the instruction of the DCSF, the IMS AS further indicates the MF to reserve separate media termination for the de-multiplexed stream in stead of using the same media termination with other multiplexed streams.

If the IMS AS is instructed by DCSF to use data channel multiplexing, when IMS AS generates SDP offer in INVITE or re-INVITE request to remote network, the DC stream IDs associated with the data channels to be multiplexed and the binding information combined with the data channels are included in the SDP offer for a single m line.

* * * * Second change * * * *

AC.7.10.3 Determination and handling of multiplexing and de-multiplexing

AC.7.10.3.1 Determination of multiplexing and de-multiplexing

The IMS AS determines that a number of data channel streams are multiplexed when it receives a (re-)INVITE request containing an SDP offer which includes multiple “a=dcmap” attributes within a single m line. When the IMS AS receives the INVITE request with data channel multiplexing, if data channel multiplexing is supported, it reports the event to the DCSF for data channel management. The IMS AS includes the media specification for each data channel application sources (e.g. a=dcmap:0, a=dcmap:10, a=dcmap:100, and/or a=dcmap:110) of the multiplexed bootstrap data channel, or the media specification for each application data channel (e.g. which is distinguished by a combination of a=dcmap line and a=3gpp-req-app line), with the same media ID via Nimsas SessionEventControl Notify as described in clause AA.2.4.2.

The DCSF determines that a number of data channel streams are multiplexed when it receives a Media info list containing multiple media specifications with the same media ID from the IMS AS. The DCSF determines whether the data channel multiplexing is allowed and whether serving network needs to de-multiplex the data channel traffic based on subscription data, binding information, the endpoints of data channels and capability of terminating network.

AC.7.10.3.2 Handling multiplexing and de-multiplexing at the originating network

The IMS AS and DCSF in the originating network follows the following principles to handle data channel multiplexing and de-multiplexing:

- If the data channels are kept multiplexed, the DCSF instructs originating IMS AS accordingly. The DCSF instructs to IMS AS by including the DC Media Specification for each application data channel stream for the single media item via Nimsas MediaControl MediaInstruction as described in clause AA.2.4.3. The originating IMS AS further instructs the MF to reserve the same media termination for multiplexed data channels via Nmf MRM services as described in clause AC.2.5.2. This attempt may succeed and IMS session establishment may be successful.
- If the data channels are kept multiplexed, the DCSF instructs originating IMS AS accordingly. The originating IMS AS further instructs the MF to reserve the same media termination for multiplexed data channels. If the terminating network rejects the SDP offer because the terminating network does not support multiplexing, then originating IMS AS and originating DCSF needs to revert to a new demultiplexed SDP offer towards the target network as described in next bullet.
- If the originating network determines to demultiplex the data channels ~~is determined~~, the DCSF in the originating network instructs originating IMS AS accordingly. The DCSF instructs to IMS AS by additionally including the associated media ID for each de-multiplexed data channel stream in Nimsas MediaControl MediaInstruction as described in clause AA.2.4.3. The originating IMS AS further instructs the MF to reserve separate media terminations for the de-multiplexed streams by including the separate media ID for each de-multiplexed stream with the same associated media ID in Nmf MRM services as described in clause AC.2.5.2. When the originating IMS AS sends the SDP offer to terminating IMS network, the data channels are not multiplexed in the SDP offer. IMS session establishment continues.
- In case of multiplexing of local bootstrap data channel and remote bootstrap data channel in originating network are kept, the DCSF instructs to originating IMS AS with the separate media ID for local bootstrap data channels and for remote bootstrap data channels with the same associated media ID, and the originating IMS AS further instructs to the MF to reserve separate media terminations for local bootstrap data channels and remote bootstrap data channels.

Editor's note: When a IMS network not supporting data channel multiplexing receives an SDP offer with multiplexed data channels for a standalone IMS DC session it rejects the session with 488. How the originating IMS network handles this scenario is FFS.

AC.7.10.3.3 Handling at the terminating network.

The IMS AS and DCSF in the terminating network follows the following principles to handle data channel multiplexing and de-multiplexing:

- If the data channels are kept multiplexed, the terminating DCSF instructs terminating IMS AS accordingly. The terminating DCSF instructs to terminating IMS AS by including the DC Media Specification for each application data channel stream for the single media item via Nimsas MediaControl MediaInstruction as described in clause AA.2.4.3. The terminating IMS AS further instructs the MF to reserve the same media termination for multiplexed data channels; via Nmf MRM services as described in clause AC.2.5.2.
- If the data channels ~~is~~are determined to be de-multiplexed, the terminating DCSF instructs terminating IMS AS accordingly. The terminating DCSF instructs to terminating IMS AS by additionally including the associated media ID for each de-multiplexed data channel stream in Nimsas MediaControl MediaInstruction as described in clause AA.2.4.3. The terminating IMS AS further instructs the MF to reserve separate media terminations for the de-multiplexed streams by including the separate media ID for each de-multiplexed stream with the same associated media ID in Nmf MRM services as described in clause AC.2.5.2. When the terminating IMS AS sends the SDP offer to terminating UE, the data channels are not multiplexed in the SDP offer.
- In case of multiplexing of local bootstrap data channel and remote bootstrap data channel in terminating network are kept, the terminating DCSF instructs to terminating IMS AS with the separate media ID for local bootstrap data channels and for remote bootstrap data channels with the same associated media ID, and the terminating IMS AS further instructs to the MF to reserve separate media terminations for local bootstrap data channels and remote bootstrap data channels.

***** End of changes *****